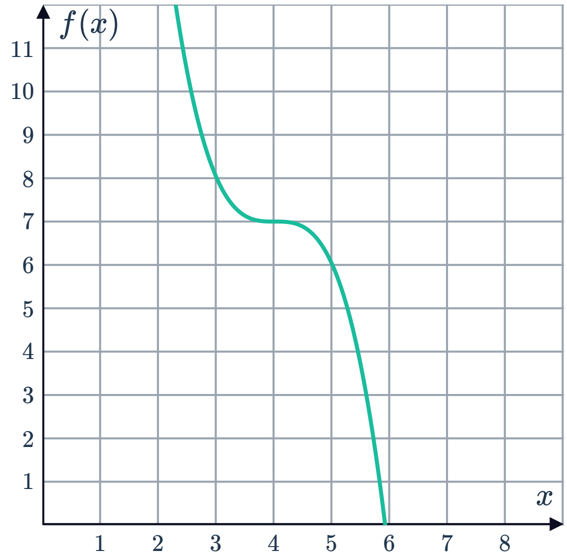
Functions and gradient functions

1 Consider the graph of the function $f(x) = -(x-4)^3 + 7$:



a State the x-value of the stationary point of $f(x)$.

b State the domain where $f(x)$ is decreasing.

2 State whether the rate of change of the following functions is positive, negative, or zero for all values of x:

a $f(x) = -8$ **b** $f(x) = 4x - 1$ **c** $f(x) = -x + 10$

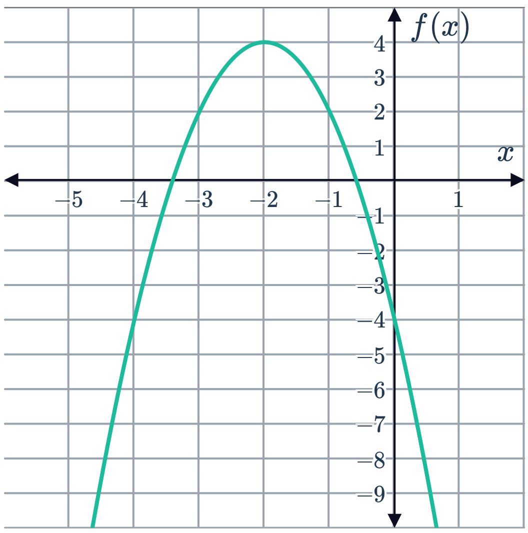
3 For each of the following functions:

i State the x-value of the stationary point of $f(x)$.

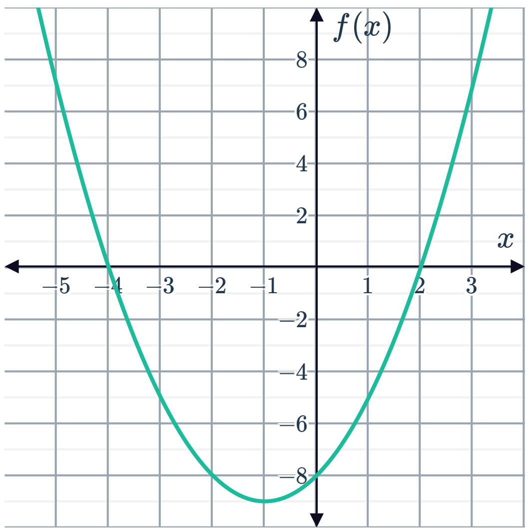
ii State the domain where $f(x)$ is increasing.

iii State the domain where $f(x)$ is decreasing.

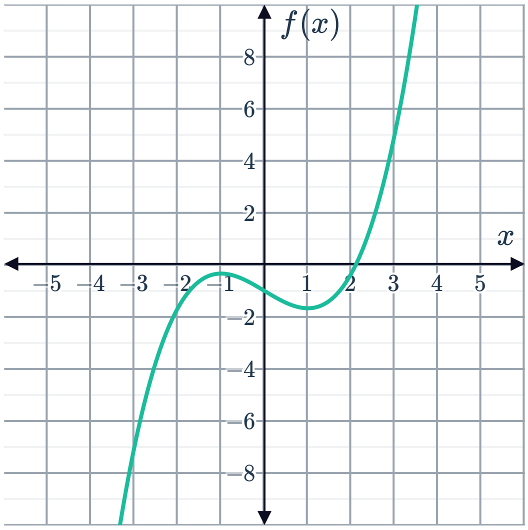
a $f(x) = -2(x+2)^2 + 4$



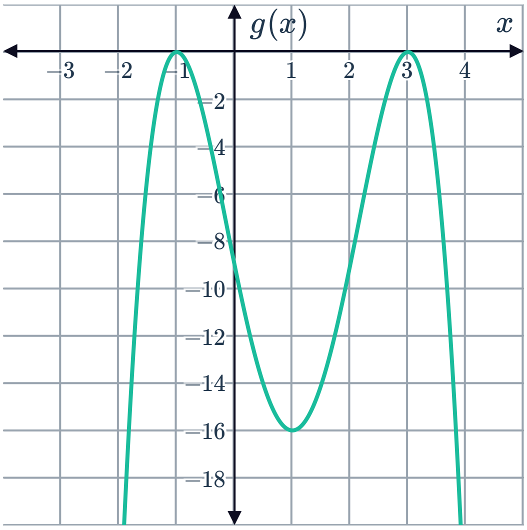
b $f(x) = (x+4)(x-2)$



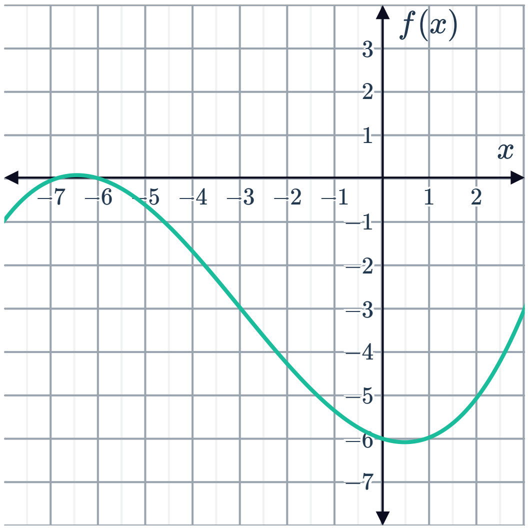
c $f(x) = \dfrac{x^3}{3} - x - 1$



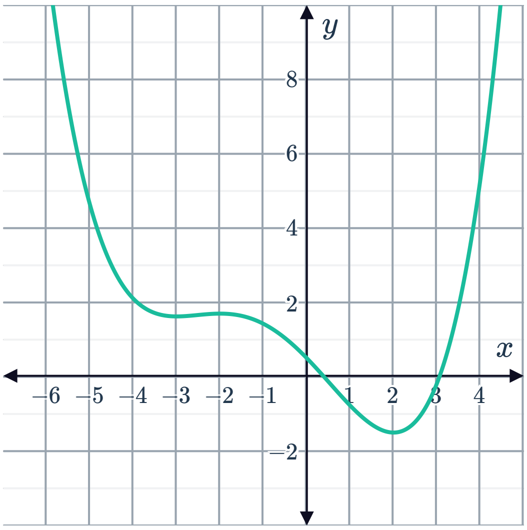
d $g(x) = -(x-3)^2(x+1)^2$



e $f(x)$



f $f(x)$



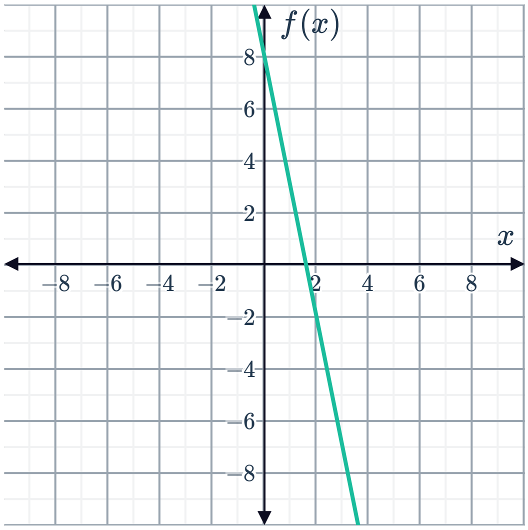
4 Consider the function $y = 4x - 3$.

a Find the gradient function.

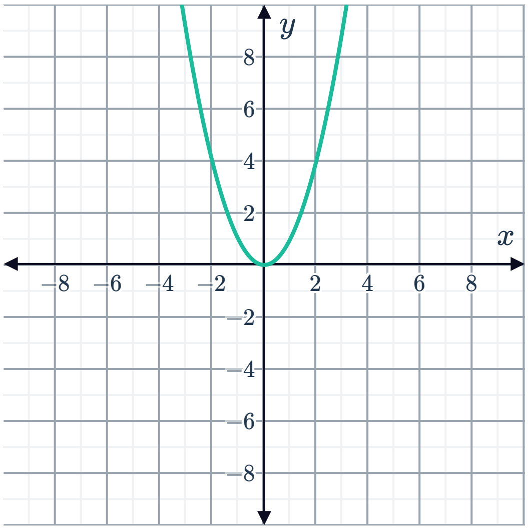
b Sketch the graph of the gradient function.

5 For each of the following functions, sketc[redacted]gradient function:

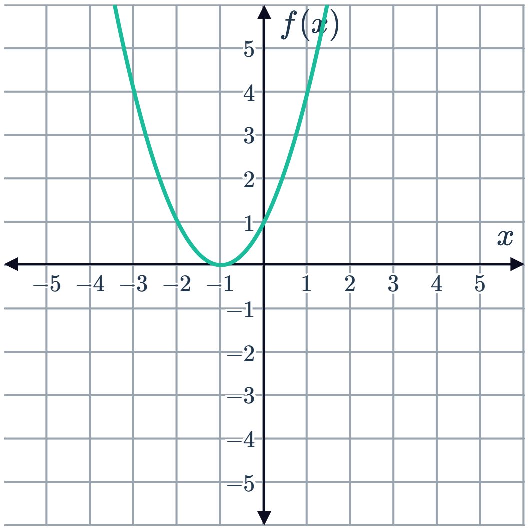
a $f(x) = -5x + 8$



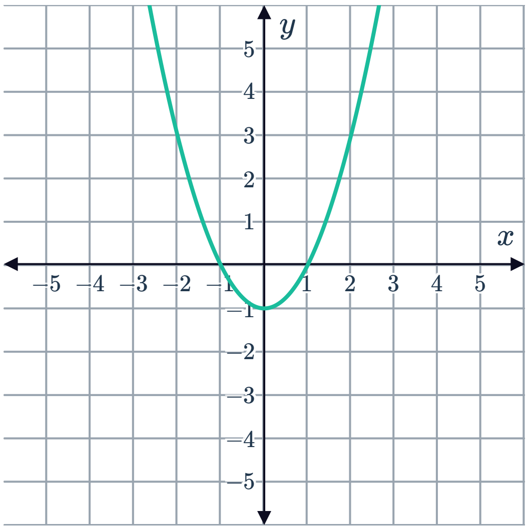
b $y = x^2$



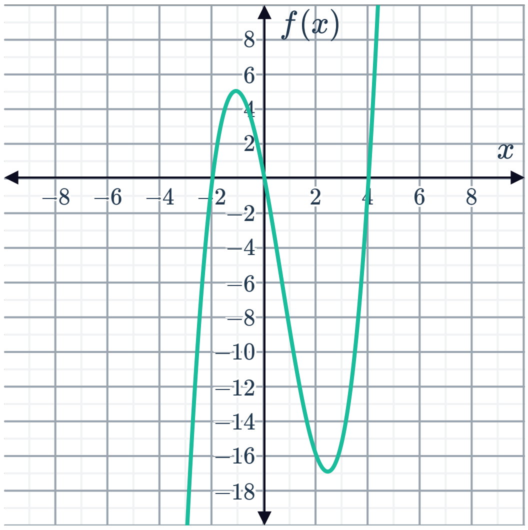
c $f(x) = (x + 1)^2$



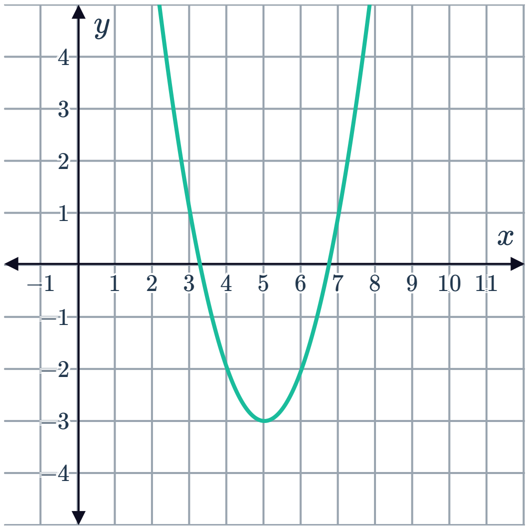
d $y = x^2 - 1$



e $f(x) = x(x - 4)(x + 2)$



6 Consider the function $y = (x-5)^2 - 3$ gra below:

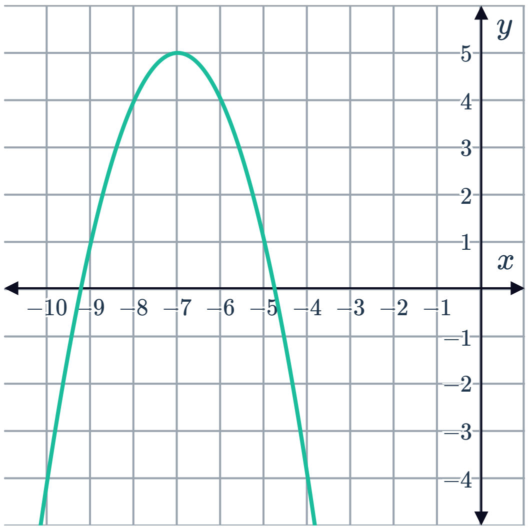


a State the x-intercept of the gradient function.

b For $x < 5$, state whether the values of the gradient function are above or below the x-axis.

c For $x > 5$, state whether the values of the gradient function are above or below the x-axis.

7 Consider the function $y = -(x+7)^2 + 5$ graphed below:

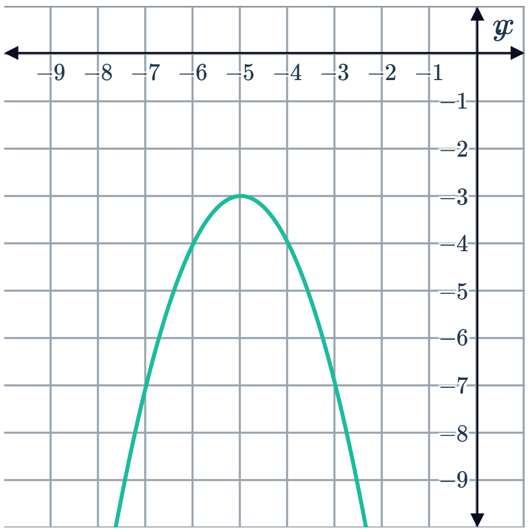


a State the x-intercept of the gradient function.

b For $x < -7$, state whether the values of the gradient function are above or below the x-axis.

c For $x > -7$, state whether the values of the gradient function are above or below the x-axis.

8 Consider the function $y = -x^2 - 10x - 28$ graphed below:

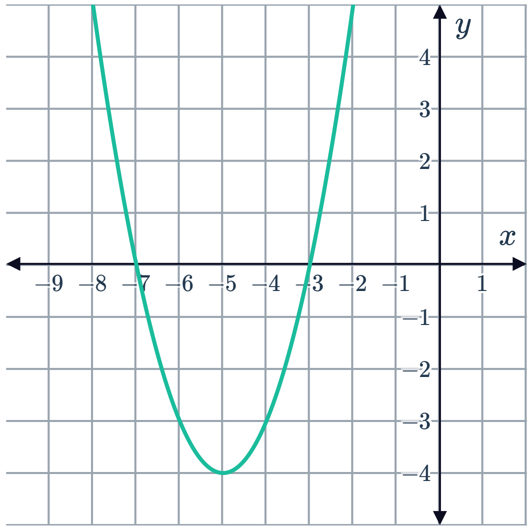


a State the x-intercept of the gradient function.

b For $x < -5$, state whether the values of the gradient function are above or below the x-axis.

c For $x > -5$, state whether the values of the gradient function are above or below the x-axis.

9 Consider the function $y = x^2 + 10x + 21$ g[redacted]ed below:

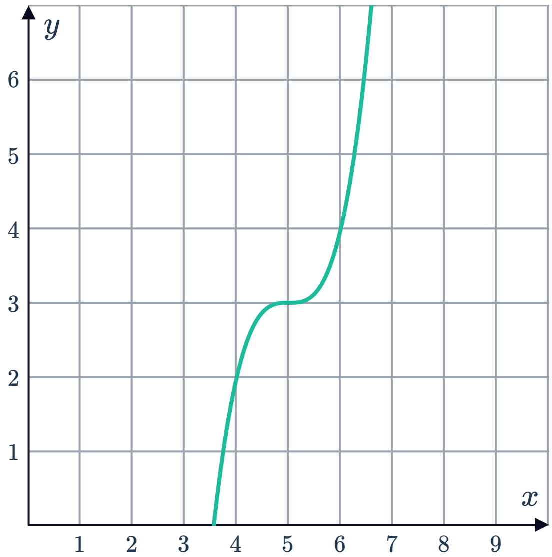


a State the x-intercept of the gradient function.

b For $x < -5$, state whether the values of the gradient function are above or below the x-axis.

c For $x > -5$, state whether the values of the gradient function are above or below the x-axis.

10 Consider the function $y = (x - 5)^3 + 3$:



a State the coordinates of the point of inflection.

b State the gradient of the curve at this point.

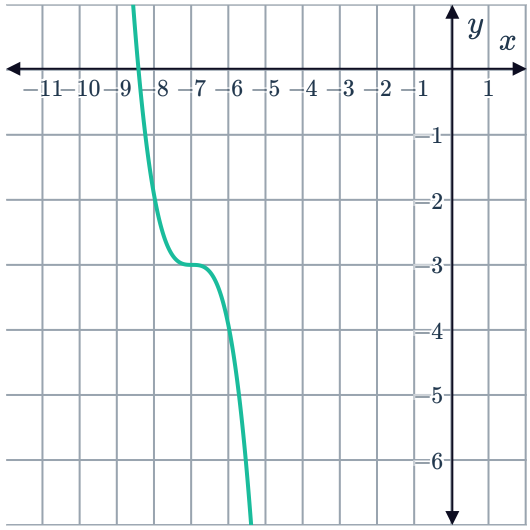
c What feature does the gradient function have at $x = 5$?

d For $x < 5$, state whether the values of the gradient function are above or below the x-axis.

e For $x > 5$, state whether the values of the gradient function are above or below the x-axis.

f What type of point is on the gradient function at $x = 5$?

11 Consider the function $y = -(x+7)^3 - 3$:



a State the coordinates of the point of inflection.

b State the gradient of the curve at this point.

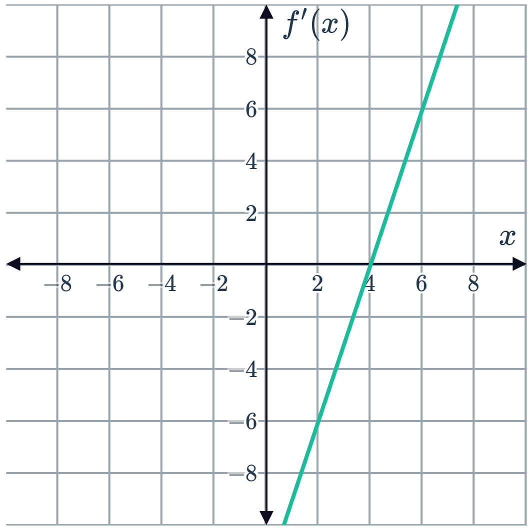
c What feature does the gradient function have at $x = -7$?

d For $x < -7$, state whether the values of the gradient function are above or below the x-axis.

e For $x > -7$, state whether the values of the gradient function are above or below the x-axis.

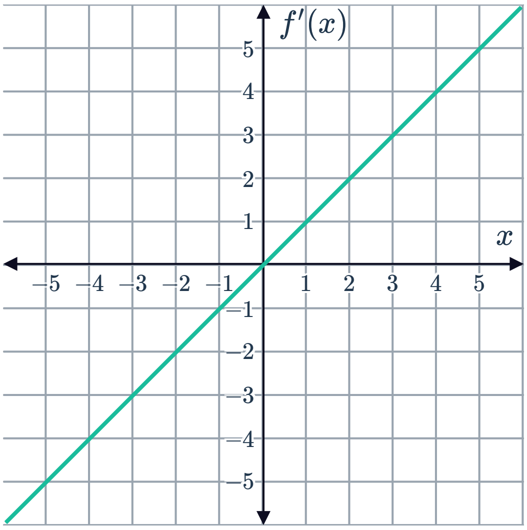
f What type of point is on the gradient function at $x = -7$?

12 Consider the graph of the gradient function $f'(x)$:
What can be said about the graph of $f(x)$?

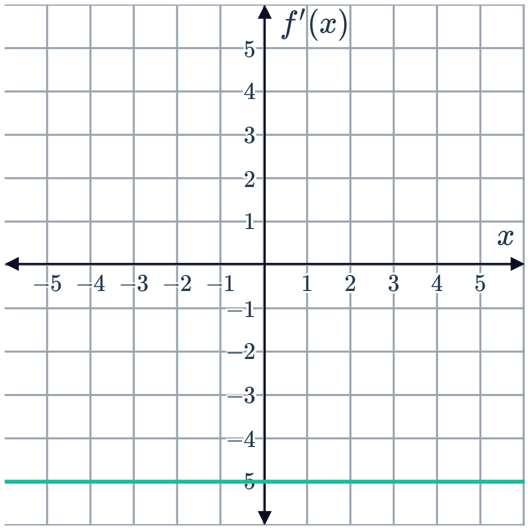


13 For each of the following gradient functio[redacted]etch a graph of a possible original function:

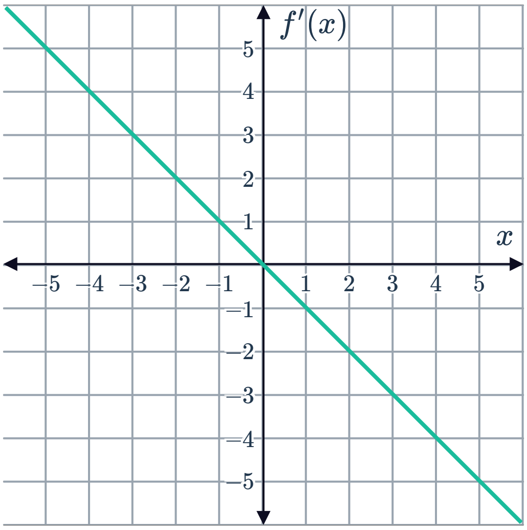
a



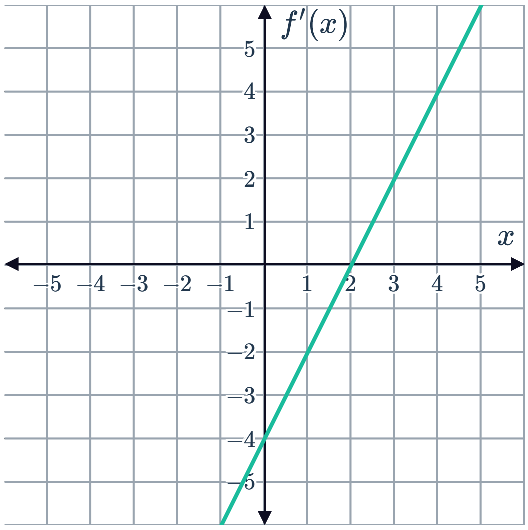
b



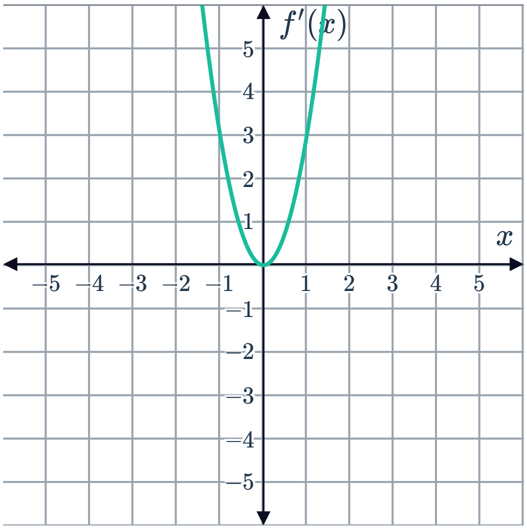
c



d



e



Types of stationary points

14 State the type of point that matches the following descriptions:

- **a** A point where the curve changes from decreasing to increasing.
- **b** A point where the curve changes from increasing to decreasing.
- **c** A point where the tangent is horizontal and the concavity changes about the point.

15 For each of the following functions:

- **i** Find the derivative.
- **ii** Find the coordinates of any stationary points.
- **iii** Classify each stationary point.

- **a** $y = -6x^2 + 84x - 29$
- **b** $y = x^3 - 21x^2 + 144x - 19$
- **c** $f(x) = (x+3)^2 (x+6)$
- **d** $f(x) = (x+5)^3 + 4$
- **e** $f(x) = -\frac{x^3}{3} + \frac{13x^2}{2} - 30x + 10$
- **f** $f(x) = (4x+5)(x+1)$
- **g** $f(x) = 134 - 300x + 240x^2 - 64x^3$
- **h** $f(x) = \left(x^2 - 9\right)^2 + 4$

16 Consider the parabola with equation $y = 5 + x - x^2$.

- **a** Find the coordinates of the vertex of the parabola.
- **b** State the gradient of the tangent to the parabola at the vertex.
- **c** What type of stationary point is at the vertex of this parabola?

17 Consider the function $f(x) = x^2 + 4x + 9$.

- **a** Find an equation for the gradient function $f'(x)$.
- **b** State the interval in which the function is increasing.
- **c** State the interval in which the function is decreasing.
- **d** Find the coordinates of the stationary point.
- **e** Classify the stationary point.

18 Consider the function $f(x) = 4x^3 + 5x^2 -$ [illegible]

a Find an equation for the gradient function $f'(x)$.

b Find the coordinates of the stationary points.

c Complete the table of values:

x	-2	$-\frac{5}{6}$	$-\frac{1}{2}$	0	1
$f'(x)$		0		0	

d Hence determine the:

i Local minimum

ii Local maximum

e Is -4 the absolute minimum value of the function? Explain your answer.

19 Consider the function $f(x) = 3x^2 - 54x + 241$.

a Find $f'(x)$.

b Find the x-coordinate of the stationary point.

c Classify the stationary point.

Graphs of functions

20 Sketch the linear function for which $f(0) = 1$ and $f'(2) = 3$.

21 Sketch the quadratic function, $f(x)$, that satisfies the following conditions:

a
- $f(0) = -18$
- $f(3) = 0$
- $f(6) = 6$
- $f'(6) = 0$
- $f'(x) > 0$ for $x < 6$

b
- $f(0) = 16$
- $f(2) = 0$
- $f(5) = -9$
- $f'(5) = 0$
- $f'(x) < 0$ for $x < 5$

c
- $f(0) = 5$
- $f(-2) = 0$
- $f'(3) = 0$
- $f'(x) > 0$ for $x < 3$

d
- $f(0) = 10$
- $f(-2) = 0$
- $f(-6) = -8$
- $f'(-6) = 0$
- $f'(x) < 0$ for $x < -6$

22 Sketch the cubic function, $f(x)$, that satisfies the following conditions:

a

- $f(0) = 7$
- $f(-2) = 0$
- $f(-4) = -1$
- $f'(-4) = 0$
- $f'(x) > 0$ for $x < -4$
- $f'(x) > 0$ for $x > -4$

b

- $f'(2) = 0$
- $f'(-3) = 0$
- $f'(x) < 0$ for $-3 < x < 2$
- $f'(x) > 0$ elsewhere

23 Sketch the quartic function, $f(x)$, that satisfies the following conditions:

a

- $f'(-1) = 0$
- $f'(4) = 0$
- $f'(x) > 0$ for $x > 4$
- $f'(x) < 0$ elsewhere

b

- $f(0) = 0$
- $f'(0) = 0$
- $f'(2) = 0$
- $f'(-2) = 0$
- $f'(x) > 0$ for $x < -2, 0 < x < 2$
- $f'(x) < 0$ elsewhere

24 Consider the equation of the parabola $y = 3x^2 - 18x + 24$.

a Find the x-intercepts.

b Find the y-intercept.

c Find $\dfrac{dy}{dx}$.

d Find the stationary point.

e Classify the stationary point.

f Sketch the graph of the parabola.

25 For each of the following functions:

i Find the y-intercept.

ii Find the x-intercepts.

iii Find $f'(x)$.

iv Hence find the x-coordinates of the stationary points.

v Classify the stationary points.

vi Sketch the graph of the function.

a $f(x) = 9x^2 + 18x - 16$

b $f(x) = (4x+5)^2 (x-1)$

c $f(x) = (2x-1)^2 (1-x)$

d $f(x) = (x+2)^3 - 1$

e $f(x) = x^3 + 11x^2 + 24x$

f $f(x) = \left(x^2 - 4\right)^2 + 4$

g $f(x) = 2(x-1)^3 - 16$

h $f(x) = x^4 - 4x^3$

i $f(x) = x^3 + 5x^2$

j $f(x) = x^2(x-3)^2$

26 Sketch the graph of the following functions, showing all stationary points:

a $y = 2x^3 - 12x^2 + 18x - 8$

b $y = 3x^5 - 20x^3 + 6$

c $y = 2x^3 + 3x^2 - 36x + 5$

d $y = 20 + 4x^3 - x^4$

e $y = -3x^4 + 16x^3 - 24x^2 + 30$

f $y = x^3 - 6x^2 - 15x + 7$